# Learning how to split

## Cognitive bias approach to the acquisition of disjunction

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# Disjunctions

- Inclusive vs exclusive disjunction

  (1) Maria ate an apple or a banana
  - a. M ate at least one of the two [inclusive]
  - b. M did not eat both [exclusive]

- Standard derivation of exclusive reading via alternative-based reasoning: [Grice 75, Horn 72, ect]

  (2) Maria ate an apple or a banana
  - a. Literal meaning: $\alpha \vee \beta$ [incl]
  - b. ALT: $\alpha \wedge \beta$
  - c. Negation of ALT: $\neg(\alpha \wedge \beta)$ [excl]

- Adults tend to prefer exclusive interpretations, but inclusive readings remain possible [Nicolae et al., 2024]

- **How do children interpret disjunction?**

# The acquisition of disjunction

## Noveck, 2001; Chierchia et al., 2001

- Experimental findings
  Children do not derive scalar implicatures, including exclusive readings
- Explanation: Ability to perform alternative-based reasoning develops late
- Conclusion: Children are more logical than adults[1]

- **Are children more logical than adults?**

[1] Noveck, 2001. When children are more logical than adults: experimental investigations of scalar implicature. *Cognition*, 78(2):165–188.

# Are children more logical than adults?

- Chierchia et al. (2001) tested acceptability of disjunction in conjunctive situations

  (3) Maria ate an apple **or** a banana

| | adults | children |
|---|---|---|
| [🍎 🍌] | no | yes |
| [🍎] | yes | ? |
| [🍌] | yes | ? |

- Singh et al. (2016), Tieu et al. (2017) investigated the remaining cases and found that many children systematically interpret disjunctions conjunctively

| | adults | children |
|---|---|---|
| [🍎 🍌] | no | yes |
| [🍎] | yes | **no** |
| [🍌] | yes | **no** |

[cf. Skordos *et al* 2020; Cochard 2025; Bleotu *et al* 2025; but also Evans 93]

- **Why do children interpret disjunction conjunctively?**

# Null hypothesis

## Hypothesis 1: Lexical Misanalysis

Children are genuinely confused between 'or' and 'and', as they play the same syntactic role.

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**Problems**

1. Children correctly reason about disjunction in some environments [Paglierini et al., 2018; Su, 2014]

   (4) Ann did **not** eat apples **or** bananas
   - $\not\rightsquigarrow$ Ann did not eat apples and bananas $\neg(\alpha \wedge \beta)$
   - $\rightsquigarrow$ Ann did not eat apples and she did not eat bananas $\neg(\alpha \vee \beta)$

2. Children can be forced to access disjunctive meanings when evaluating incompatible disjuncts [Bleotu et al., 2024]

   (5) The squirrel is at the top or at the bottom of the tree
   - $\not\rightsquigarrow$ The squirrel is at the top and at the bottom of the tree

# More findings: NOT OR = NOT AND

- Children (and many adults) interpret conjunctions under negation as disjunction [homogeneity, Sbardolini 2023]

(6) Maria did **not** eat apples **and** bananas
$\not\rightsquigarrow$ At least one kind of fruit was not eaten by Maria $\neg(\alpha \wedge \beta)$
$\rightsquigarrow$ Maria did not eat apples & she did not eat bananas $\neg(\alpha \vee \beta)$

- Children across languages have strong 'neither' interpretations for both NOT OR & NOT AND sentences irrespective of the scopal preferences prescribed by their language

| *Language* | NOT OR | NOT AND | |
|---|---|---|---|
| English | $\neg(\alpha \vee \beta)$ | $\neg(\alpha \wedge \beta)$ | |
| Mandarin | $\neg\alpha \vee \neg\beta$ | $\neg\alpha \wedge \neg\beta$ | (Notley, Zhou & Crain, 2016) |
| English (child) | $\neg\alpha \wedge \neg\beta$ | $\neg\alpha \wedge \neg\beta$ | |
| Mandarin (child) | $\neg\alpha \wedge \neg\beta$ | $\neg\alpha \wedge \neg\beta$ | (Notley, Zhou et al., 2012, 2016) |
| Italian (child) | $\neg\alpha \wedge \neg\beta$ | $\neg\alpha \wedge \neg\beta$ | (Goro, 2007; Geçkin et al., 2016) |
| Turkish (child) | $\neg\alpha \wedge \neg\beta$ | $\neg\alpha \wedge \neg\beta$ | (Goro, 2007; Geçkin et al., 2016) |
| Japanese (child) | $\neg\alpha \wedge \neg\beta$ | $\neg\alpha \wedge \neg\beta$ | (Goro & Akiba, 2004; Goro, 2007) |

Table: Adults and children interpretation of negated disjunction & conjunction

# More findings: Conjunctive free choice (CFC)

- While children have difficulties with scalar implicatures, 3 year olds easily derive free choice inferences (Tieu et al., 2016):

  (7) Standard free choice (FC)

  a. You **may** eat an apple **or** a banana $\rightsquigarrow$ you are allowed to choose between an apple and a banana

  b. $\Diamond(\alpha \vee \beta) \rightsquigarrow \Diamond\alpha \wedge \Diamond\beta$

- Cochard et al. (2024) experimentally showed that "conjunctive children" in fact derive stronger conjunctive free choice (CFC) inferences:

  (8) Conjunctive free choice (CFC)

  a. You **may** eat an apple **or** a banana $\rightsquigarrow$ you are allowed to eat both an apple and a banana

  b. $\Diamond(\alpha \vee \beta) \rightsquigarrow \Diamond(\alpha \wedge \beta)$

# Summary of children inferences

(9) OR $\Rightarrow$ AND

a. M ate an apples **or** a banana $\rightsquigarrow$ M ate an apples and a banana

b. $(\alpha \vee \beta) \rightsquigarrow (\alpha \wedge \beta)$

(10) NOT OR

a. Ann did **not** eat apples **or** bananas $\rightsquigarrow$ Ann did not eat apples and she did not eat bananas

b. $\neg(\alpha \vee \beta) \rightsquigarrow \neg\alpha \wedge \neg\beta$

(11) NOT AND

a. Ann did **not** eat apples **and** bananas $\rightsquigarrow$ Ann did not eat apples and she did not eat bananas

b. $\neg(\alpha \wedge \beta) \rightsquigarrow \neg\alpha \wedge \neg\beta$

(12) CFC

a. You **may** eat an apple **or** a banana $\rightsquigarrow$ You may eat both an apple and a banana

b. $\Diamond(\alpha \vee \beta) \rightsquigarrow \Diamond(\alpha \wedge \beta)$

# Grammatical view

## Hypothesis 2: Singh et al., 2016

Children derive conjunctive readings by alternative-based reasoning

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**Assumptions**

1. Children can perform (recursive) alternative-based reasoning;
2. Children are not aware that AND is an alternative to OR;
3. Children know the inclusive (logical) meaning of OR, but their alternative-based derivation leads to incorrect results because of (2).

# Grammatical view (Singh et al., 2016)

**Derivation** OR $\Rightarrow$ AND

(13) Ann ate an apple **or** a banana.

a. Literal meaning: $\alpha \vee \beta$ (inclusive OR)
b. Alt: $\alpha; \beta$ (and crucially not $\alpha \wedge \beta$)
c. ExhAlt: $(\alpha \wedge \neg\beta); (\beta \wedge \neg\alpha)$
d. Negation of ExhAlt: $\neg(\alpha \wedge \neg\beta); \neg(\beta \wedge \neg\alpha)$ $(\equiv \alpha \leftrightarrow \beta)$
e. $\rightsquigarrow$ Ann ate **both** an apple and a banana

Singh et al. (2016) propose that children know the meaning of disjunction, but use it only to go through a complex recursive process to generate conjunctive readings

**Derivation** NOT OR = NOT AND

(14) Ann did **not** eat apples **or/and** bananas.

a. Literal meaning: $\neg(\alpha \vee \beta)$ / $\neg(\alpha \wedge \beta)$
b. Alt: $\neg\alpha; \neg\beta$
c. ExhAlt: $(\alpha \wedge \neg\beta); (\beta \wedge \neg\alpha)$
d. Negation of ExhAlt: $\neg(\alpha \wedge \neg\beta); \neg(\beta \wedge \neg\alpha)$ $(\equiv \alpha \leftrightarrow \beta)$
e. $\rightsquigarrow$ Ann ate **neither** an apple **nor** a banana

**Why would children go through these complicated processes just to arrive at logically incorrect readings?**

# Grammatical view (Singh et al., 2016)

- Derivations triggered by application of EXH: a covert grammatical operator which can freely occur in various positions

(15) a. $\text{EXH}(\alpha \vee \beta) \Rightarrow \alpha \wedge \beta$ YES
b. $\alpha \vee \beta \not\Rightarrow \alpha \wedge \beta$ NO

- In the negation cases, EXH crucially must scope above negation

(16) a. $\text{EXH}\neg(\alpha \vee \beta)$ / $\text{EXH}\neg(\alpha \wedge \beta) \Rightarrow \neg\alpha \wedge \neg\beta$ YES
b. $\neg\text{EXH}(\alpha \vee \beta)$ / $\neg\text{EXH}(\alpha \wedge \beta) \not\Rightarrow \neg\alpha \wedge \neg\beta$ NO

- To derive conjunctive free choice inferences, however, EXH must occur under the modal

(17) a. $\text{EXH}\Diamond(\alpha \vee \beta) \not\Rightarrow \Diamond(\alpha \wedge \beta) \Rightarrow \Diamond\alpha \wedge \Diamond\beta$ NO
b. $\Diamond\text{EXH}(\alpha \vee \beta) \Rightarrow \Diamond(\alpha \wedge \beta)$ YES

**How and why would children systematically and across languages choose these specific grammatical configurations?**

- Semantic Subset Maxim (SSM)/Strongest Meaning Hypothesis (SMH): children prefer strong readings
- But SSM/SMH descriptive rather than explanatory unless we assume that children are by nature perfect reasoners sensitive to logical strength

# Cognitive bias approach to the acquisition of disjunction

Conjunctive readings: not the result of complicated processes / rational choices but rather arise because children are still developing some cognitive abilities

## Our proposal

Children who interpret disjunctions conjunctively do so because they have not yet fully developed:

(i) the ability to engage with models that verify sentences by virtue of an empty configuration (**zero-models**);

(ii) the ability to process (conflicting) alternatives in a parallel fashion (**split-states**).



Klochowicz



Sbardolini

Klochowicz, Tomasz, Giorgio Sbardolini, and MA (2026). "Cognitive bias approach to the acquisition of disjunction". In: *Proceedings of Sinn und Bedeutung. Vol. 30*, pp. 554–571. https://doi.org/10.18148/sub/2026.v30.1353.

# Cognitive bias approach (NIHIL$^+$)

Common assumption: Reasoning and understanding of natural language involve the creation of mental models [Johnson-Laird 1983, Mascarenhas & Koralus 2013, a.o.]

- **Understanding** a sentence S means being able to mentally construct a model picturing the world which verifies S, and possibly also a model which falsifies it
- **Reasoning** depends on two main processes: first construct verifying models for the premises and then check the validity of the conclusion on these models

NIHIL$^+$ hypothesis: biases can constrain the construction of these models and therefore impact both reasoning and interpretation:

- NEGLECT-ZERO: a tendency to avoid emptiness
- NO-SPLIT: a negative bias towards the parallel processing of multiple alternatives
- ...

The proposal:

- Children (age 3-5/6) more sensitive to these biases than adults
- Combination of neglect-zero and no-split explain children 'conjunctive' interpretations

# Neglect-zero

NEGLECT-ZERO
when interpreting a sentence speakers construct models depicting reality (some verifying the sentence, some falsifying it) $\mapsto$ common assumption
and in this process tend to neglect models that verify the sentence by virtue of an empty configuration (*zero-models*) $\mapsto$ new hypothesis




- Tendency to neglect zero-models follows from the cognitive difficulty of:
  1. conceiving emptiness, the absence of things rather than their presence
  2. evaluating truths with respect to empty witness sets

[Nieder 2016; O. Bott *et al* 2019]

# Neglect-zero

## Illustration

(18) Less than three squares are black.

a. Verifier: [■, □, ■]
b. Falsifier: [■, ■, ■]
c. Zero-models: [□, □, □]; [■, ■, ■]; . . .

Zero-models in (18-c) verify the sentence by virtue of an empty set of black squares

- Cognitive difficulty of zero-models confirmed by experimental findings and connected to / can be argued to explain:
  - the special status of 0 among the natural numbers [Nieder 2016]
  - why downward-monotonic quantifiers are more costly to process than upward-monotonic ones (*less* vs *more*) [O. Bott *et al* 2019]
- NZ hypothesis: neglect-zero also at the origin of many common departures from classical reasoning
  1. Free choice & ignorance: A or B ⇒ it might be A & it might be B [MA 2022]
  2. Existential Import: every A is B ⇒ some A is B
  3. Principles from connexive logic: [Ziegler, Knudstorp & MA 2026]
     - Aristotle's Theses: if not A then A ⇒ ⊥; if A then not A ⇒ ⊥
     - Boethius' Thesis: if A then B & if A then not B ⇒ ⊥
  4. . . .

# Neglect-zero effects on disjunction

## Illustrations

(19) Maria ate an apple.

a. Verifier: [🍎]

b. Falsifiers: [🍌]; [🍐]; [ ]

c. Zero-models: none

(20) Maria ate a banana.

a. Verifier: [🍌]

b. Falsifiers: [🍎]; [🍐]; [ ]

c. Zero-models: none

(21) M ate an apple and a banana.

a. Verifier: [🍎 🍌]

b. Falsifiers: [🍐]; [ ]

c. Zero-models: none

(22) M ate an apple or a banana.

a. Verifier: ?

b. Falsifiers: [🍐]; [ ]

c. Zero-models: ?

# Novel hypothesis: neglect-zero effects on disjunction

## Illustrations

(23) Maria ate an apple.

a. Verifier: [🍎]

b. Falsifiers: [🍌]; [🍐]; [ ]

c. Zero-models: none

(24) Maria ate a banana.

a. Verifier: [🍌]

b. Falsifiers: [🍎]; [🍐]; [ ]

c. Zero-models: none

(25) M ate an apple and a banana.

a. Verifier: [🍎 🍌]

b. Falsifiers: [🍐]; [ ]

c. Zero-models: none

(26) M ate an apple or a banana.

a. Verifier: ?

b. Falsifiers: [🍐]; [ ]

c. **Zero-models**: [🍎]; [🍌]

- Two **zero-models** in (26-c): verify the sentence by virtue of an empty witness for one of the disjuncts

# Novel hypothesis: neglect-zero effects on disjunction

## Illustrations

(27) Maria ate an apple.

a. Verifier: [🍎]

b. Falsifiers: [🍌]; [🍐]; [ ]

c. Zero-models: none

(28) Maria ate a banana.

a. Verifier: [🍌]

b. Falsifiers: [🍎]; [🍐]; [ ]

c. Zero-models: none

(29) M ate an apple and a banana.

a. Verifier: [🍎 🍌]

b. Falsifiers: [🍐]; [ ]

c. Zero-models: none

(30) M ate an apple or a banana.

a. Verifier: [🍎 | 🍌] ⇐ 'split'

b. Falsifiers: [🍐]; [ ]

c. **Zero-models**: [🍎]; [🍌]

- Two **zero-models** in (30-c): sentence verified by virtue of an empty witness for one of the disjuncts
- **Split state** in (30-a): multiple (possibly conflicting) alternatives processed in a parallel fashion
- Neglect-zero hypothesis: Zero-models, where only one of the disjuncts is depicted, are cognitively taxing and therefore disregarded
- The exclusion of zero-models from the set of verifiers for a disjunction leads directly to an account of ignorance inferences and FC

# Back to the conjunctive children

(31) Maria ate an apple or a banana.

a. Verifier: [🍎 | 🍌] [⇐ **split state**]

b. Falsifiers: [🍐]; [ ]

c. **Zero-models**: [🍎]; [🍌]

## Hypothesis 3: Klochowicz, Sbardolini & MA, 2026

Children have conjunctive readings as they (similarly to adults) neglect zero and, unlike adults, lack the ability to split

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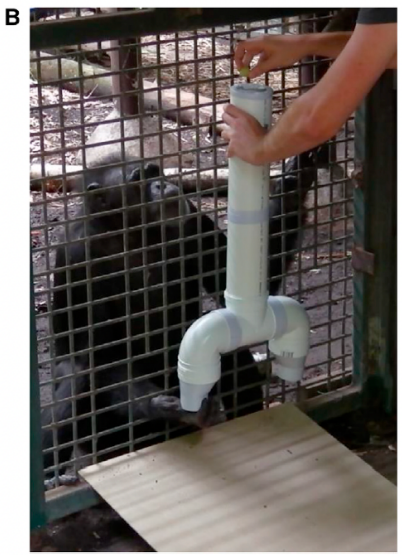
- **Split states**: multiple (possibly conflicting) alternatives processed in a parallel fashion $\mapsto$ also a cognitively taxing operation
- We conjecture that the ability to split is acquired late
- The combination of neglect-zero and no-split can explain conjunctive interpretations observed in preschoolers

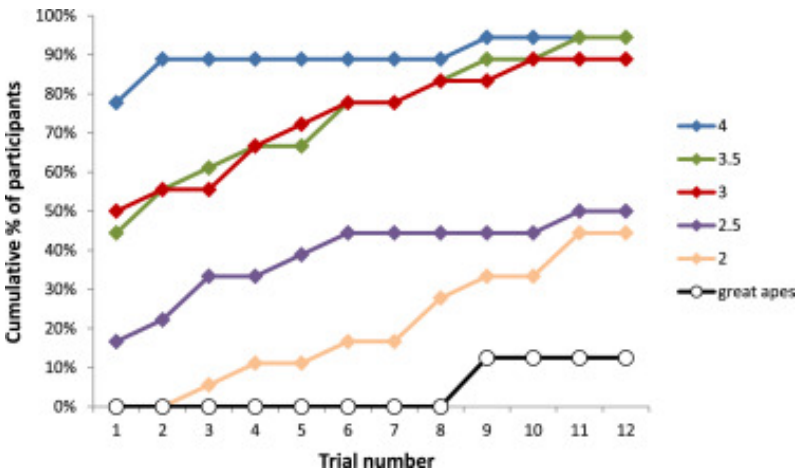
# The new bias: no-split

**NO-SPLIT**
Children (and sometimes adults) have difficulties with the parallel processing of multiple possibly conflicting alternatives (**split-states**)

## Experimental evidence




Children (age 2-4y) and apes have trouble conceiving possible different futures
(Redshaw & Suddendorf 2016)

# Our derivation of conjunctive readings

(32) Maria ate an apple or a banana

1. Deriving ignorance:

   Apple OR banana $\Rightarrow_{\text{NZ}}$ 🍎 + 🍌 $\Rightarrow_{\text{SPLIT}}$ [🍎 | 🍌]

   $\rightsquigarrow$ It might be an apple and it might be a banana (adults)

2. Deriving conjunctive reading:

   Apple OR banana $\Rightarrow_{\text{NZ}}$ 🍎 + 🍌 $\Rightarrow_{\text{NO-SPLIT}}$ [🍎 🍌]

   $\rightsquigarrow$ Both an apple and a banana (children)

3. In case of incompatible alternatives: [Leahy & Carey 2020, Bleotu *et al*, 2024]

   Left OR right $\Rightarrow_{\text{NZ}}$ ↙ + ↘ $\Rightarrow_{\text{NO-SPLIT}}$ **contradiction** ($\bot$)

   $\rightsquigarrow$ Sequential or disjunctive interpretations (children)

# Modelling biases in team semantics

## Team semantics

- Formulas interpreted wrt a set of points of evaluation (a team) rather than single ones [Hodges 1997; Väänänen 2007]
  - Classical modal logic: $[M = \langle W, R, V \rangle]$

$$M, w \models \phi, \text{ where } w \in W$$

  - Team-based modal logic:

$$M, t \models \phi, \text{ where } t \subseteq W$$

- Two crucial features
  - The empty set is among the possible teams ($\emptyset \subseteq W$) $\mapsto$ zero-models
  - Multi-membered teams can model parallel processing of alternatives $\mapsto$ split states

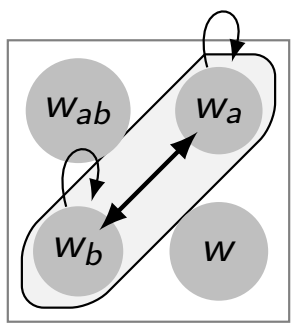



Figure: A multi-membered team modelling the parallel processing of possibilities $a$ & $b$

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## Neglect-zero & no-split

- Model-theoretic method: biases modelled by disallowing empty (neglect-zero) and multi-membered teams (no-split)
- Syntactic method: biases modelled via new logical atoms/operators
  - Neglect-zero: via **non-emptiness atom** NE which disallows empty teams as possible verifiers [Yang & Väänänen 2017]

$$M, t \models \text{NE} \text{ iff } t \neq \emptyset$$

  - No-split: via **flattening operator** F which induces pointwise evaluations and therefore mimics ban on parallel processing of alternatives

$$M, t \models \text{F}\phi \text{ iff for all } w \in t : M, \{w\} \models \phi$$

# Modelling biases in team semantics

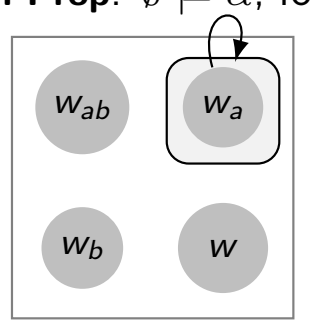
## Classical and enriched interpretations

- $\alpha \Rightarrow$ empty team & split allowed $\mapsto$ **classical (literal) meaning**
- $[\alpha]^\bullet \Rightarrow$ empty team not allowed $\mapsto$ **NZ-enriched**
- $[\alpha]^{\bullet\bullet} \Rightarrow$ neither empty team nor split allowed $\mapsto$ **NZ+NS-enriched**

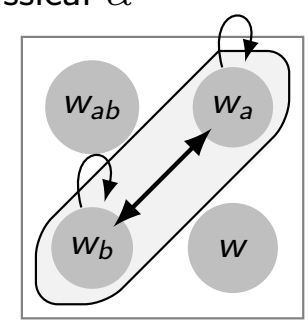
## Tensor disjunction in team semantics

- Team $t$ supports a **disjunction** iff $t$ is the union of two subteams, each supporting one of the disjuncts
  - $a \vee b$: empty team allowed $\mapsto$ subteams can be empty (**classical**)
  - $[a \vee b]^\bullet$: empty team not allowed $\mapsto$ subteams cannot be empty (**NZ-enr**)
  - $[a \vee b]^{\bullet\bullet}$: neither empty team nor split allowed $\mapsto$ no empty or distinct subteams allowed (**NZ+NS-enriched**)

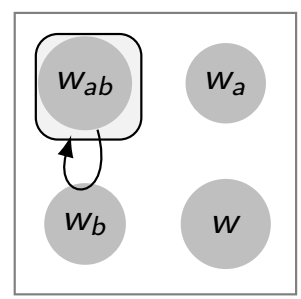
**Empty Team Prop**: $\emptyset \models \alpha$, for all classical $\alpha$ [**PROP**$= \{a, b\}$; $W = \{w_{ab}, w_a, w_b, w\}$]




$\models a \vee b$
$\not\models [a \vee b]^\bullet$
$\not\models [a \vee b]^{\bullet\bullet}$
$\mapsto$ **Zero-model**




$\models a \vee b$
$\models [a \vee b]^\bullet$
$\not\models [a \vee b]^{\bullet\bullet}$
$\mapsto$ **No-zero**, **split**




$\models a \vee b$
$\models [a \vee b]^\bullet$
$\models [a \vee b]^{\bullet\bullet}$
$\mapsto$ **No-zero**, **no-split**

# BSML$^{\mathrm{F}}$: Classical Modal Logic + NE + F

## Language

$$\phi := p \mid \mathrm{NE} \mid \neg\phi \mid \phi \vee \phi \mid \phi \wedge \phi \mid \Diamond\phi \mid \mathrm{F}\phi$$

## Bilateral team semantics

$[M = \langle W, R, V\rangle \;\&\; s, t, t' \subseteq W]$

$$\begin{array}{rll}
M, s \models p & \text{iff} & \text{for all } w \in s : V(w, p) = 1 \\
M, s =\!| p & \text{iff} & \text{for all } w \in s : V(w, p) = 0 \\
M, s \models \neg\phi & \text{iff} & M, s =\!| \phi \\
M, s =\!| \neg\phi & \text{iff} & M, s \models \phi \\
M, s \models \phi \vee \psi & \text{iff} & \text{there are } t, t' : t \cup t' = s \;\&\; M, t \models \phi \;\&\; M, t' \models \psi \\
M, s =\!| \phi \vee \psi & \text{iff} & M, s =\!| \phi \;\&\; M, s =\!| \psi \\
M, s \models \phi \wedge \psi & \text{iff} & M, s \models \phi \;\&\; M, s \models \psi \\
M, s =\!| \phi \wedge \psi & \text{iff} & \text{there are } t, t' : t \cup t' = s \;\&\; M, t =\!| \phi \;\&\; M, t' =\!| \psi \\
M, s \models \Diamond\phi & \text{iff} & \text{for all } w \in s : \text{there is } t \subseteq R[w] : t \neq \emptyset \;\&\; M, t \models \phi \\
M, s =\!| \Diamond\phi & \text{iff} & \text{for all } w \in s : M, R[w] =\!| \phi \\
M, s \models \mathrm{NE} & \text{iff} & s \neq \emptyset \qquad [\text{where } R[w] = \{v \in W \mid wRv\}] \\
M, s =\!| \mathrm{NE} & \text{iff} & s = \emptyset \\
M, s \models \mathrm{F}\phi & \text{iff} & \text{for all } w \in s : M, \{w\} \models \phi \\
M, s =\!| \mathrm{F}\phi & \text{iff} & \text{for all } w \in s : M, \{w\} =\!| \phi
\end{array}$$

**Entailment:** $\phi_1, \ldots, \phi_n \models \psi$ iff for all $M, s$: $M, s \models \phi_1, \ldots, M, s \models \phi_n \Rightarrow M, s \models \psi$

# Some facts

- F collapse in NE-free fragment. If $\alpha$ is NE-free,

$$M, s \models \mathrm{F}\alpha \quad \text{iff} \quad \text{for all } w \in s : M, \{w\} \models \alpha \quad \text{iff} \quad M, s \models \alpha$$
$$M, s \not\models \mathrm{F}\alpha \quad \text{iff} \quad \text{for all } w \in s : M, \{w\} \not\models \alpha \quad \text{iff} \quad M, s \not\models \alpha$$

- NE-free fragment is already flat, i.e., equivalent to classical modal logic. If $\alpha$ is NE-free,

$$\alpha \models_{BSML} \beta \text{ iff } \alpha \models_{ML} \beta \qquad [\text{if } \alpha, \beta \text{ are NE-free}]$$

- But in interaction with NE, flattening yields non-trivial effects

# Implementations: two neglect-zero enrichment functions

- Two ways to model neglect-zero in BSML: [MA 2022]
  1. $[\,]^+$: enriches formulas with the requirement to satisfy NE distributed along each of their subformulas $\mapsto$ BSML$^+$
  2. $[\,]^*$: equivalent to ruling out $\emptyset$ from the set of possible teams $\mapsto$ BSML$^*$

- For NE- and $\mathrm{F}$-free $\alpha$,

  1. $[\alpha]^+$ defined as follows: [MA 2022]

$$\begin{aligned}
[p]^+ &= p \wedge \mathrm{NE} \\
[\neg\alpha]^+ &= \neg[\alpha]^+ \wedge \mathrm{NE} \\
[\alpha \vee \beta]^+ &= ([\alpha]^+ \vee [\beta]^+) \wedge \mathrm{NE} \\
[\alpha \wedge \beta]^+ &= ([\alpha]^+ \wedge [\beta]^+) \wedge \mathrm{NE} \\
[\Diamond\alpha]^+ &= \Diamond[\alpha]^+ \wedge \mathrm{NE}
\end{aligned}$$

  2. $[\alpha]^*$ defined as follows: [Klochowicz 2026]

$$\begin{aligned}
[p]^* &= (p \wedge \mathrm{NE}) \vee \neg\mathrm{NE} \\
[\neg\alpha]^* &= \neg([\alpha]^* \wedge \mathrm{NE}) \vee \neg\mathrm{NE} \\
[\alpha \vee \beta]^* &= (([\alpha]^* \vee [\beta]^*) \wedge \mathrm{NE}) \vee \neg\mathrm{NE} \\
[\alpha \wedge \beta]^* &= (([\alpha]^* \wedge [\beta]^*) \wedge \mathrm{NE}) \vee \neg\mathrm{NE} \\
[\Diamond\alpha]^* &= (\Diamond[\alpha]^* \wedge \mathrm{NE}) \vee \neg\mathrm{NE}
\end{aligned}$$

[$^*$-implementation $\equiv$ ruling out $\emptyset$ from the set of possible teams]

# Implementations: two neglect-zero enrichment functions

- Both implementations derive possibility & free choice inferences:
  - Possibility: $[\alpha \vee \beta]^{+/*} \models \Diamond_e \alpha \wedge \Diamond_e \beta$ [$R$ is state-based]
  - Free choice: $[\Diamond(\alpha \vee \beta)]^{+/*} \models \Diamond\alpha \wedge \Diamond\beta$
- But different behaviour under negation:

  $\neg[p]^+ = \neg(p \wedge \mathrm{NE}) = \neg p \vee \neg\mathrm{NE} = \neg p \vee \bot = \neg p$

  $\neg[p]^* = \neg((p \wedge \mathrm{NE}) \vee \neg\mathrm{NE}) = \neg(p \wedge \mathrm{NE}) \wedge \neg\neg\mathrm{NE} = (\neg p \vee \neg\mathrm{NE}) \wedge \mathrm{NE} = \neg p \wedge \mathrm{NE}$
- Only BSML$^*$ derives negative possibility and negative free choice:
  - Negative possibility: $[\neg(\alpha \wedge \beta)]^* \models \Diamond_e \neg\alpha \wedge \Diamond_e \neg\beta$ [$R$ is state-based]
  - Negative FC: $[\neg\Box(\alpha \wedge \beta)]^* \models \Diamond\neg\alpha \wedge \Diamond\neg\beta$

# Implementations: adding no-split

- Implementation of no-split: uses flattening operator $\mathrm{F}$

$$M, s \models \mathrm{F}\phi \text{ iff for all } w \in s : M, \{w\} \models \phi$$

  Flattening $\mapsto$ formulas always interpreted wrt to singleton states
- Facts:
  1. OR $\Rightarrow$ AND: $\mathrm{F}[\alpha \vee \beta]^{+/*} \models \alpha \wedge \beta$
  2. Conjunctive FC: $\Diamond\mathrm{F}[\alpha \vee \beta]^{+/*} \models \Diamond(\alpha \wedge \beta)$ $[\not\models \Box(\alpha \wedge \beta)]$
  3. NOT OR: $\neg\mathrm{F}[\alpha \vee \beta]^{+/*} \models \neg\alpha \wedge \neg\beta$
  4. NOT AND: $\neg\mathrm{F}[\alpha \wedge \beta]^{*} \models \neg\alpha \wedge \neg\beta$, but $\neg\mathrm{F}[\alpha \wedge \beta]^{+} \not\models \neg\alpha \wedge \neg\beta$
- Both BSML$^+$ and BSML$^*$ derive:
  - $\mapsto$ conjunctive *or* in combination with flattening (1-2)
  - $\mapsto$ classical disjunctive interpretation under negation (3)
- But only BSML$^*$ predicts
  - 'Neither' readings for NOT AND in combination with $\mathrm{F}$ (4) [Sbardolini 2023]
- Only in BSML$^+$, $\emptyset$ is needed as part of the building blocks

  $\Rightarrow$ Natural to assume BSML$^*$ for "conjunctive" children who plausibly do not yet have full access to $\emptyset$

# Implementations: no-split enrichment

For NE- and F-free $\alpha$, $[\alpha]^{**}$ defined as follows:

$$\begin{aligned}
[p]^{**} &= (\mathrm{F}p \wedge \mathrm{NE}) \vee \neg\mathrm{NE} \\
[\neg\alpha]^{**} &= (\mathrm{F}\neg[\alpha]^{**} \wedge \mathrm{NE}) \vee \neg\mathrm{NE} \\
[\alpha \vee \beta]^{**} &= (\mathrm{F}([\alpha]^{**} \vee [\beta]^{**}) \wedge \mathrm{NE}) \vee \neg\mathrm{NE} \\
[\alpha \wedge \beta]^{**} &= (\mathrm{F}([\alpha]^{**} \wedge [\beta]^{**}) \wedge \mathrm{NE}) \vee \neg\mathrm{NE} \\
[\Diamond\alpha]^{**} &= (\mathrm{F}\Diamond[\alpha]^{**} \wedge \mathrm{NE}) \vee \neg\mathrm{NE}
\end{aligned}$$

## Predictions

1. OR $\Rightarrow$ AND: $[\alpha \vee \beta]^{**} \models \alpha \wedge \beta$
2. Conjunctive FC: $\Diamond[\alpha \vee \beta]^{**} \models \Diamond(\alpha \wedge \beta)$ $\quad [\not\models \Box(\alpha \wedge \beta)]$
3. NOT OR: $\neg[\alpha \vee \beta]^{**} \models \neg\alpha \wedge \neg\beta$
4. NOT AND: $\neg[\alpha \wedge \beta]^{**} \models \neg\alpha \wedge \neg\beta$

# Two views on the acquisition of disjunction

- Grammatical view: 'conjunctive' children can compute implicatures but do not have access to lexical scalar alternatives ⟨or, and⟩;
- NIHIL$^+$: conjunctive behaviour derives from the combination of two cognitive biases: no-zero & no-split.

| | **conjunctive** *or* | **inclusive** *or* | **exclusive** *or* |
|---|---|---|---|
| Grammatical | exh-alt [✓]<br>scalar-alt [no] | no alternatives | scalar-alt [✓]<br>(exh-alt [✓]) |
| NIHIL$^+$ | zero [no]<br>split [no] | split [✓]<br>(or zero [✓] ) | split [✓] & scalar reasoning [✓] |

- Two different predicted orders of acquisition/simplicity:
  - Grammatical view:

    (a) inclusive *or* < conjunctive *or* < exclusive *or*

  - NIHIL$^+$:

    (b) conjunctive *or* < inclusive *or* < exclusive *or*

- Preliminary experimental evidence for (b): [Bleotu et al., 2025]
  - Conjunction is children's default interpretation of a connective
  - There are more conjunctive children in the younger age groups
- Need for future longitudinal study

# Conclusions

- Conjunctive interpretations of disjunction: a mismatch between logic and language
- Grammatical implicature view:
  - Conjunctive *or* derived by recursive application of covert grammatical operator EXH in absence of conjunctive alternatives
- Nihil$^+$ proposal: Conjunctive meanings consequence of not yet fully developed cognitive abilities
  - Conjunctive *or* as no-zero + no-split effect
- Implementation in BSML$^{\mathrm{F}}$ (a bilateral team-based modal logic)
  - Literal meanings (NE-free fragment) + cognitive factors (NE, F) $\Rightarrow$ conjunctive *or*
- Need for further (longitudinal) studies

THANK YOU![2]

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